Flight Price Prediction

Model development, validation, and submission

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Github Link: https://github.com/rajataswani/mlAcadsBeyondDimension

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Executive Summary

This project develops a regression model to predict flight ticket prices from flight and itinerary information. The training data contains 18,076 rows and the test data contains 4,520 rows. I checked data quality, compared linear and tree-boosting models, tested a log transformation of the target, engineered route and flight-number features, and evaluated a blend of XGBoost and LightGBM.

The strongest candidate was a 50/50 blend of log-target XGBoost and LightGBM. On a second shuffled five-fold cross-validation run, the blend achieved a pooled out-of-fold MAE of 42.69 and RMSE of 122.88. The resulting full-training-data predictions were saved in a candidate submission CSV with the same 4,520 IDs and the same two columns as the supplied template. These validation scores estimate performance; they are not the competition's private-test score.

1. Problem and Objective

The task is supervised regression: use the labeled training rows to learn a mapping from flight attributes to the numeric ticket-price target, then predict prices for the unlabeled test rows. The practical objective is to build a reproducible pipeline that generalizes to unseen rows and can be explained in a report and viva. Public leaderboard performance is only one part of the stated evaluation; preprocessing, model variety, and explanation also matter.

2. Data Description and Quality Checks

The training file contains 18,076 rows and 12 columns, including the target price. The test file contains 4,520 rows and 11 feature columns. The submission template contains 4,520 rows and the columns id and price. The feature set includes eight categorical columns (airline, flight, source city, departure time, stops, arrival time, destination city, and flight class), two measured numeric columns (duration and days left), and an ID column.

Check	Training data	Test data
Rows	18,076	4,520
Missing cells	0	0
Duplicate rows	0	0
Duplicate IDs	0	0
IDs shared across train and test	0	0

The target price has mean 280.26, median 234, standard deviation 221.97, minimum 0, and maximum 8,996. The distribution is right-skewed, with a small number of much higher prices. The zero and high-price observations were retained rather than removed because no verified data error was established.

Average prices varied across ID ranges, but train and test contained similar proportions of IDs across those ranges. A direct model comparison showed that adding ID did not improve validation scores, so ID was omitted from the selected models. This treats the ID as an identifier rather than assuming that it represents time or another known cause.

3. Preprocessing and Feature Engineering

No missing-value imputation was needed because the audit found no missing cells. The categorical features were handled according to the model: LightGBM received categorical columns directly, while XGBoost and Ridge used one-hot encoding. One-hot encoding creates indicator columns for observed categories and ignores any unseen category during prediction.

The target price was nonnegative, including a minimum of zero. For selected models, I trained on log1p(price), which compresses the right tail and is defined for zero. Predictions were converted back to the original price scale using expm1 before evaluation. This allowed all reported errors to remain interpretable in the original price units.

Two candidate features were added: a directional route formed by combining source and destination city, and a numeric flight number parsed from the flight code. With log-target XGBoost, the original features scored MAE 44.76 and RMSE 124.42. Route alone scored 43.83 and 124.09; flight number alone scored 44.51 and 123.85; both together scored 43.66 and 123.54. Because the combination performed best in that comparison, both features were retained.

4. Validation and Evaluation Metrics

I used shuffled five-fold cross-validation. In each fold, four-fifths of the labeled rows were used to fit a model and the remaining fifth to evaluate predictions. Each training row therefore received an out-of-fold prediction from a model that had not trained on that row. The initial split used random_state=42. A second shuffled five-fold run used random_state=2026 to check whether the model ranking persisted under a different fold assignment.

MAE is the average absolute difference between actual and predicted price; it is easy to interpret in the target's units and is less sensitive to a few very large errors. RMSE is the square root of mean squared error; it penalizes large misses more strongly. Both were reported because the target contains a high-price tail and the official competition metric was not included in the materials reviewed for this report.

5. Model Comparison

The first five-fold out-of-fold comparison showed that log-target boosting models outperformed the log-target Ridge baseline. The log transformation improved XGBoost on both metrics. Adding route and flight-number features improved it further. The table reports pooled out-of-fold metrics from the same comparison run.

Model / features	MAE	RMSE
Ridge, log target	71.37	147.55
XGBoost, raw target	48.01	129.57
XGBoost, log target	44.76	124.42
XGBoost, log + route + flight number	43.66	123.54
LightGBM, log + engineered features	44.02	124.46
50/50 XGBoost + LightGBM blend	42.59	122.53

The blend weights were explored using the first run's out-of-fold predictions. The 50/50 blend had the lowest MAE in the tested grid; the 60/40 XGBoost/LightGBM blend had a nearly identical RMSE (122.49 versus 122.53). I selected the simple 50/50 blend as a balanced candidate because it achieved the lowest MAE and nearly the lowest RMSE, then checked it on a second fold assignment.

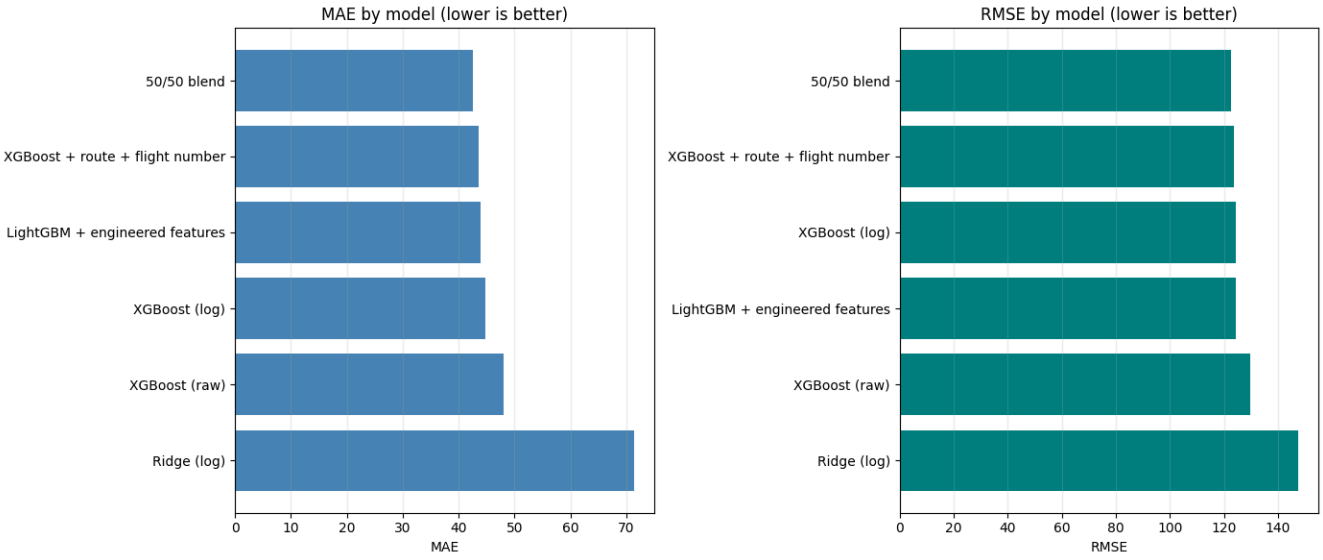


Figure 1. Pooled out-of-fold MAE and RMSE for model variants in the first shuffled five-fold run. Lower values indicate smaller errors.

6. Repeat-Split Check and Final Candidate

On the second shuffled five-fold run, engineered XGBoost achieved MAE 43.83 and RMSE 125.68; engineered LightGBM achieved MAE 44.25 and RMSE 124.38; and the fixed 50/50 blend achieved MAE 42.69 and RMSE 122.88. The blend was better than either individual model on both metrics in this run. This repeat-split check supports the blend choice, but it is not an independent external test because the same dataset was reused with a different fold assignment.

After selecting the candidate, both models were retrained on all 18,076 labeled rows. Their predictions for the 4,520 test rows were averaged equally. The candidate file, submission_blended.csv, contains the expected id and price columns, 4,520 rows, and test IDs in the template order. All prices were numeric, finite, and nonnegative. The model's maximum test prediction was approximately 2,134.70.

7. Error Analysis and Limitations

Out-of-fold analysis showed that the log-target XGBoost model performed well on common low and middle price bands but tended to underpredict expensive fares. For actual prices above 1,000, the average actual value was 1,488.36 while the average prediction was 1,053.34. The most expensive training example (8,996) was also severely underpredicted in validation. The small number of high-price records makes these estimates uncertain, and no arbitrary correction was applied to final predictions.

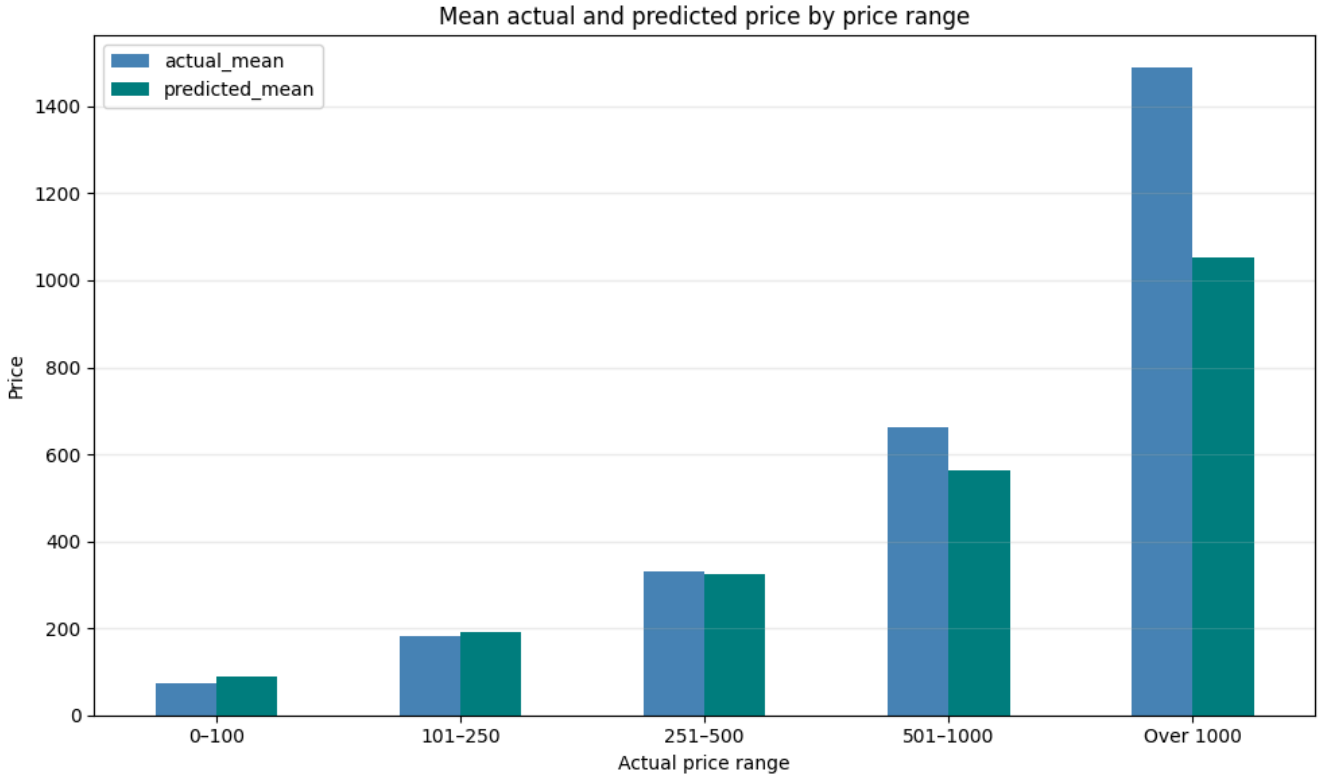


Figure 2. Mean actual and predicted price by observed-price band for baseline log-target XGBoost out-of-fold predictions. The high-price bands show underprediction.

Other limitations are that the official scoring metric was not supplied, the model settings were not exhaustively tuned, and no grouped validation by route or flight identifier was performed. Random folds assume that rows are sufficiently exchangeable for the hidden test process. If the competition splits by route, flight, or time, a grouped or time-aware validation could be more appropriate. The private-test score is unknown until evaluation is completed.

8. Conclusion

The project produced a reproducible candidate pipeline using log-target XGBoost and LightGBM with route and numeric flight-number features. A 50/50 blend had the strongest overall cross-validation results among the tested candidates and remained best on a second fold assignment. The candidate submission file passed structural checks. The main known weakness is underprediction of very expensive fares; this should be described in the report and viva rather than hidden by unsupported manual adjustments.